

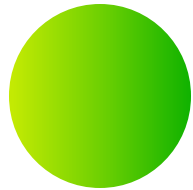
Predicting Physical Properties of Materials: A Quantum Crystallographer's Perspective

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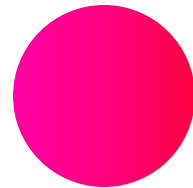
Georg-August-Universität Göttingen

Main goals



Correlation between the crystal structure and chosen physical/chemical property

Insight into electronic structure of the studied systems

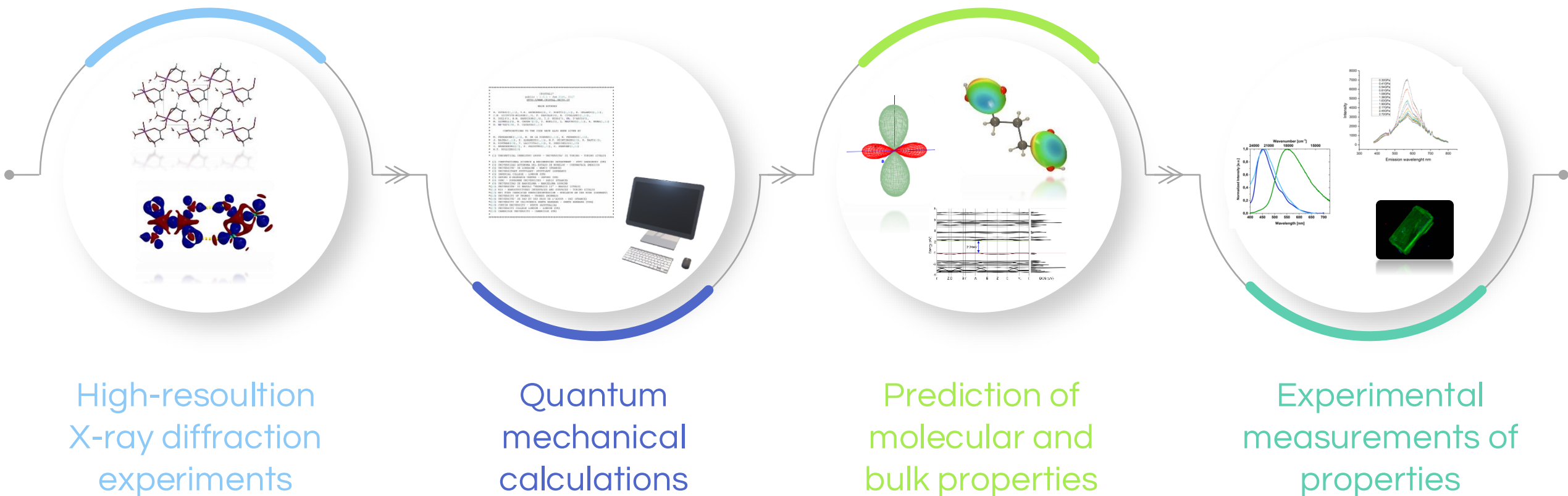


Quantification of intermolecular interaction – quantum crystallography approach

Prediction of chosen physical and chemical properties



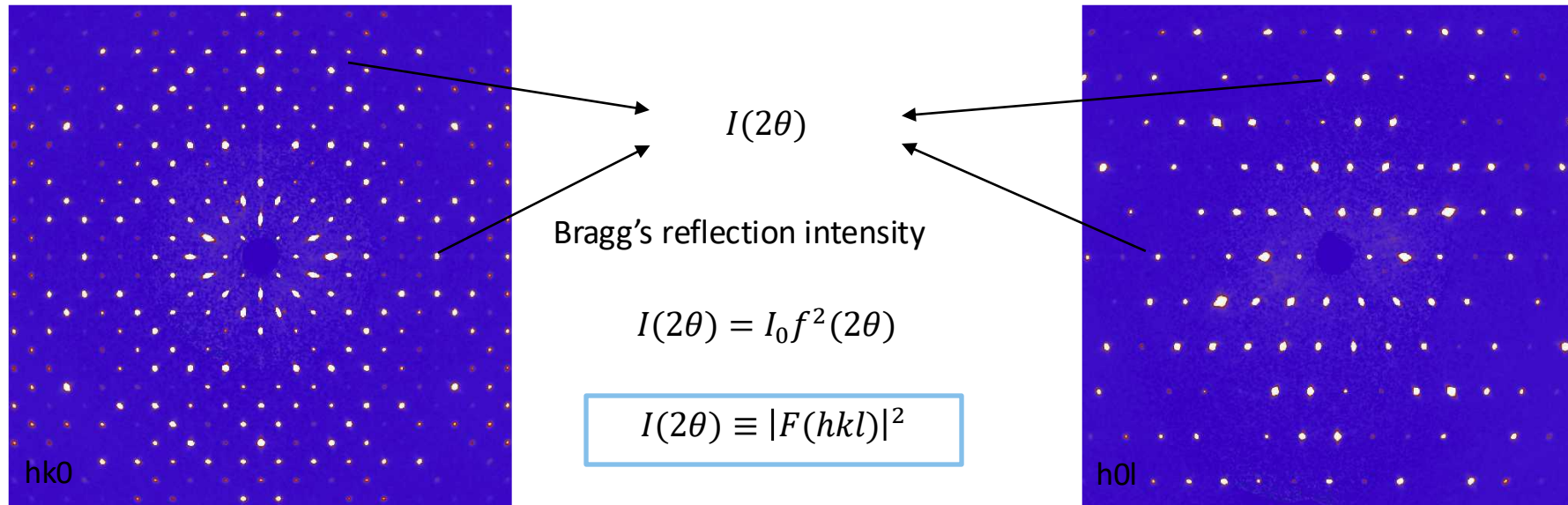
Our usual workflow



Why to engage periodic calculations?

- ❑ No access to high resolution experimental X-ray diffraction data $\Rightarrow$ *required to know electron density distribution*
- ❑ Data obtained from theory corresponds well to experimental data $\Rightarrow$ *may be „cheaper” to perform*
- ❑ No experimental measurement of a physical property available $\Rightarrow$ *we want to know anyway*
- ❑ Screening of physical properties towards efficient materials $\Rightarrow$ *informed design*

X-ray diffraction – what do we measure?



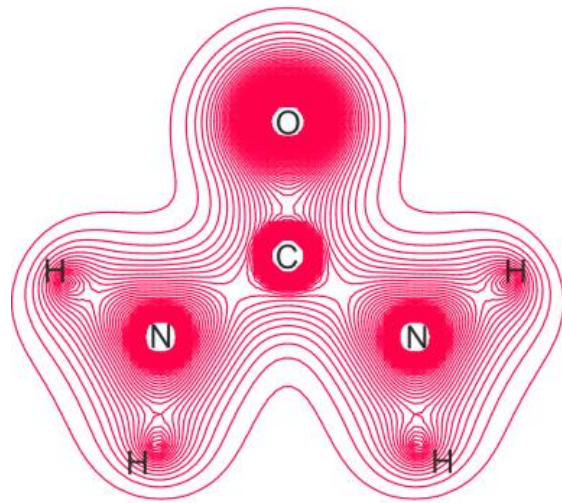
$$F(hkl) = \int_V \langle \rho(\mathbf{r}) \rangle \exp(2\pi i \mathbf{r} \cdot \mathbf{H}) = FT \langle \rho(\mathbf{r}) \rangle$$

f_j – atomic scattering factor of j atom

$F(hkl)$ – structure factor

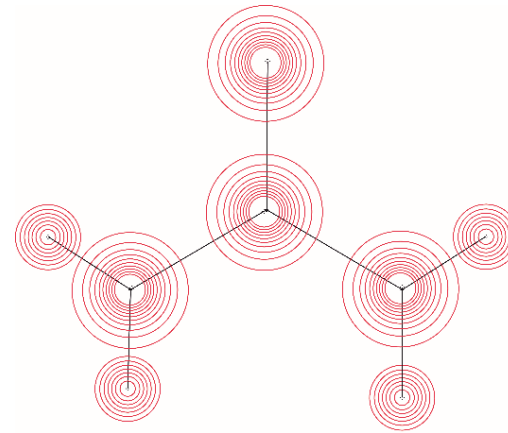
$\langle \rho(\mathbf{r}) \rangle$ – thermally averaged electron density

X-ray diffraction – what do we see?



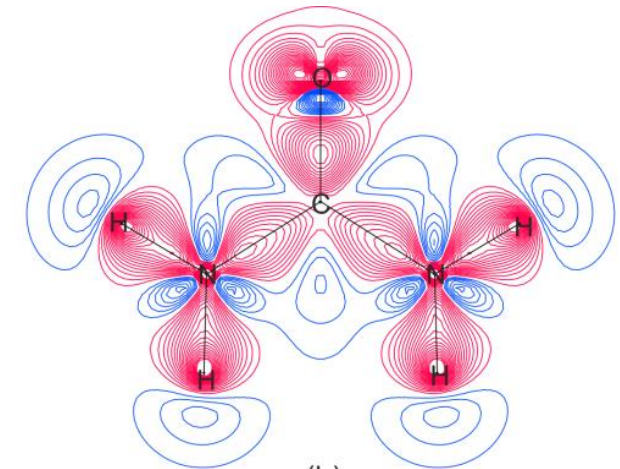
$$\rho_{obs} \equiv \rho_{total}$$

Total electron density



$$\rho_{IAM}(\mathbf{r}) \equiv \rho_{pro}$$

Conventional X-ray diffraction
experiment (IAM, independent
atom model)



$$\Delta\rho(\mathbf{r})$$

Deformation of electron density
not available via IAM

Electron density modelling – how to get the details?

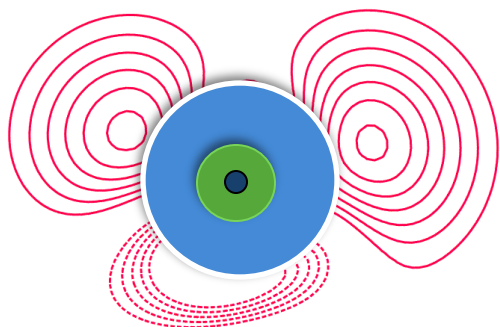
Multipole model of electron density (MM)

$$\rho_{\text{pseudoatom}} = \rho_{\text{core}} + \rho_{\text{valence}} + \rho_{\text{deformation}}$$

↑
spherical
core
density

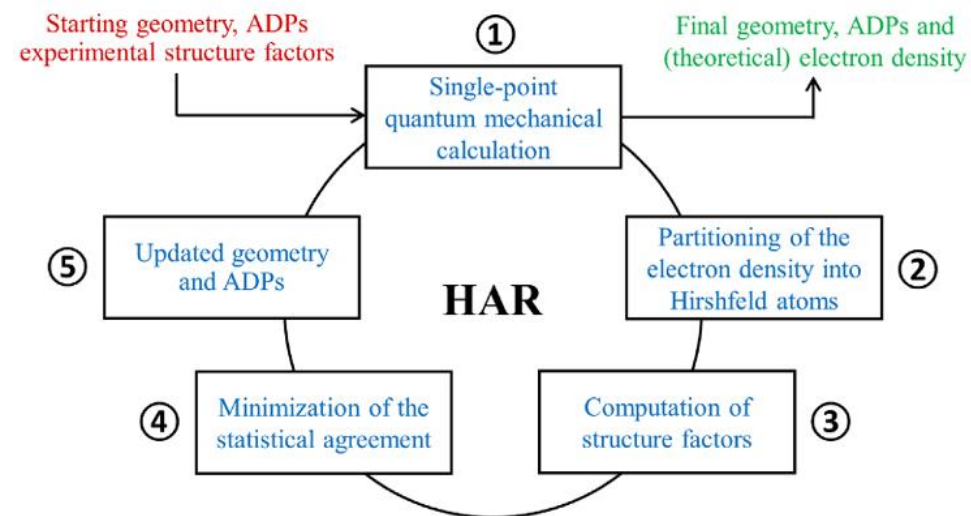
↑
spherical
valence
density

↑
aspherical
deformation
density



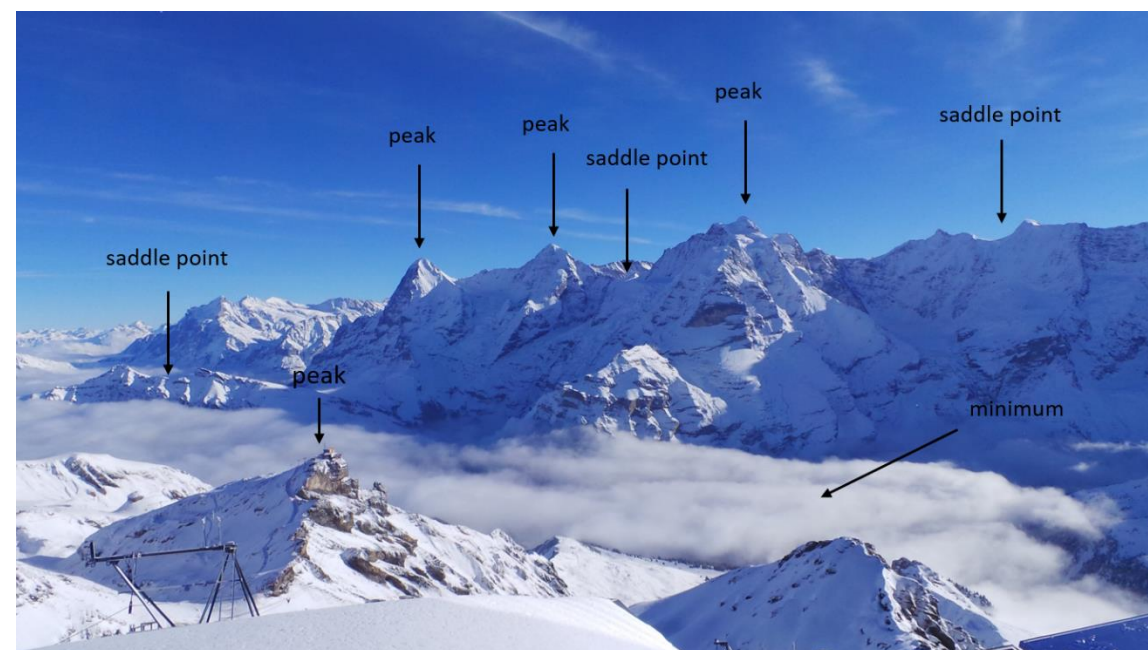
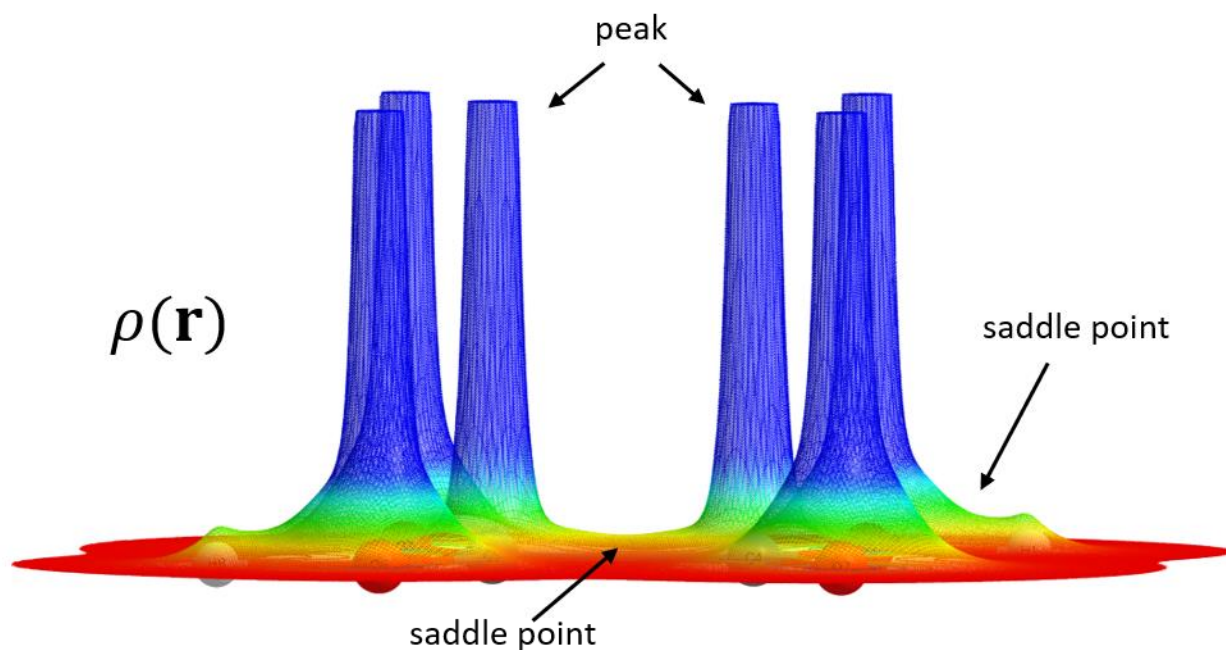
$$\begin{aligned} \rho_{\text{atom}}(\mathbf{r}) = & P_c \rho_{\text{core}}(\mathbf{r}) + P_v k^3 \rho_{\text{valence}}(k\mathbf{r}) \\ & + \sum_{l=0}^{l_{\text{max}}} \kappa'^3 R_l(\kappa'\mathbf{r}) \sum_{m=0}^l P_{lm\pm} d_{lm\pm}(\theta, \phi) \end{aligned}$$

Hirshfeld Atom Refinement (HAR)

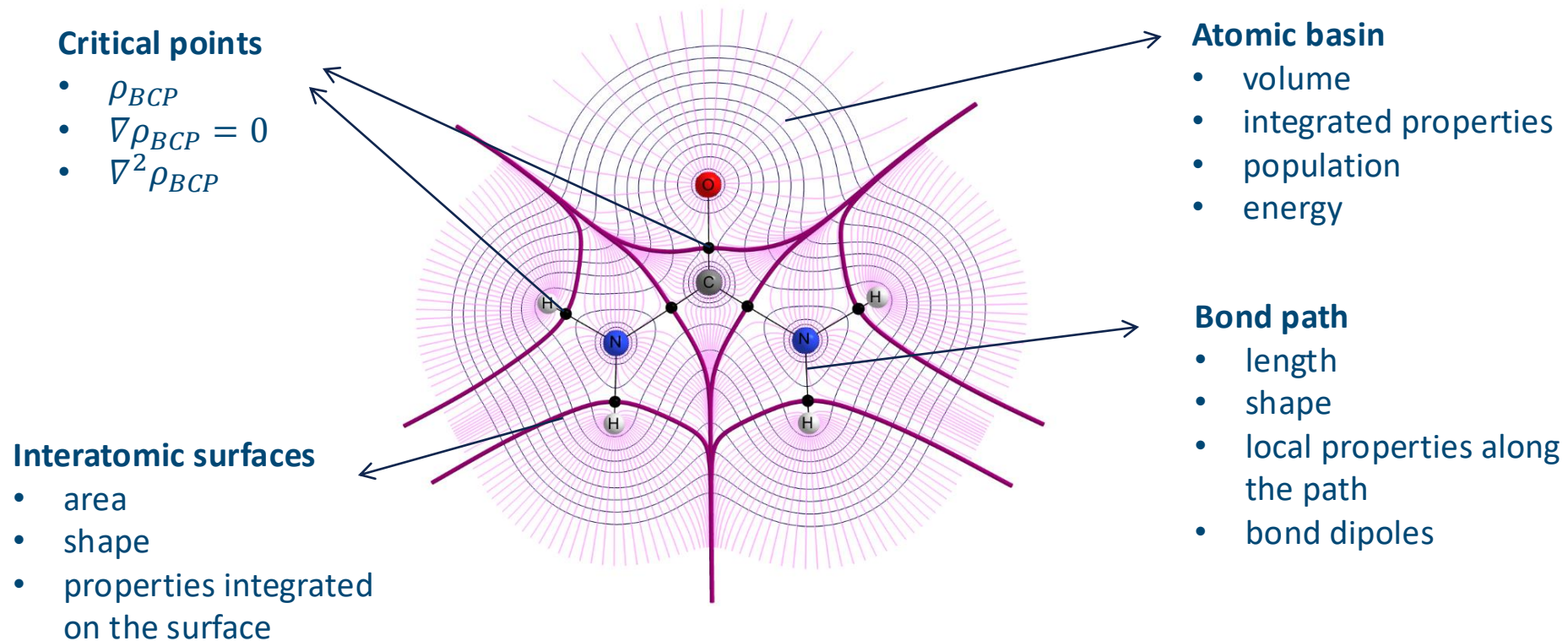


$$\chi^2 = \frac{1}{N_r - N_p} \sum_{\mathbf{H}} \frac{(\eta |F_{\mathbf{H}}^{\text{calc}}| - |F_{\mathbf{H}}^{\text{exp}}|)^2}{\sigma_{\mathbf{H}}^2}$$

Topological analysis of electron density (QTAIM)



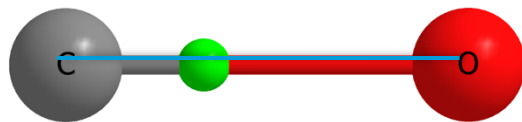
Topological analysis of electron density (QTAIM)



Topological analysis of electron density (QTAIM)

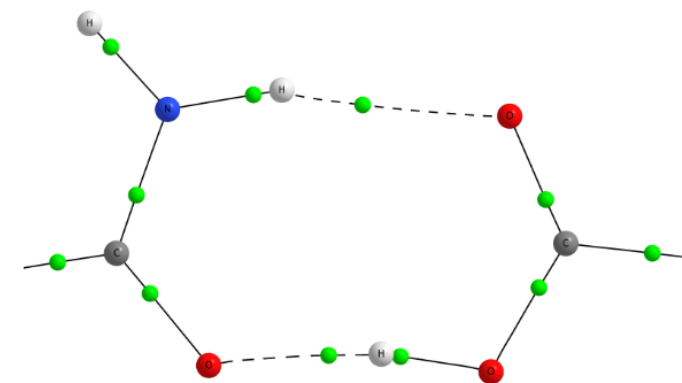
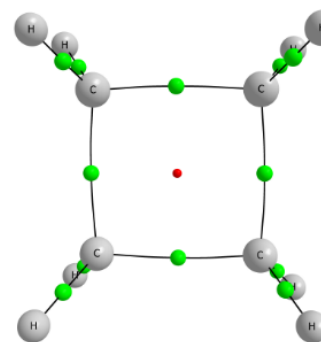
Existence of bond critical point BCP and a bond path BP between two atoms is a necessary and sufficient condition for existence of chemical bonding

- ❖ Chemical bonding not always equals to chemical bond defined with Lewis model (eg. covalent bond)
- ❖ The bond path cannot represent through bond or multi-centre interactions, but it is obviously affected by them
- ❖ Length of BP should coincide with internuclear axis (bond length)



RAB – internuclear distance

BPL – bond path length



Topological analysis of electron density

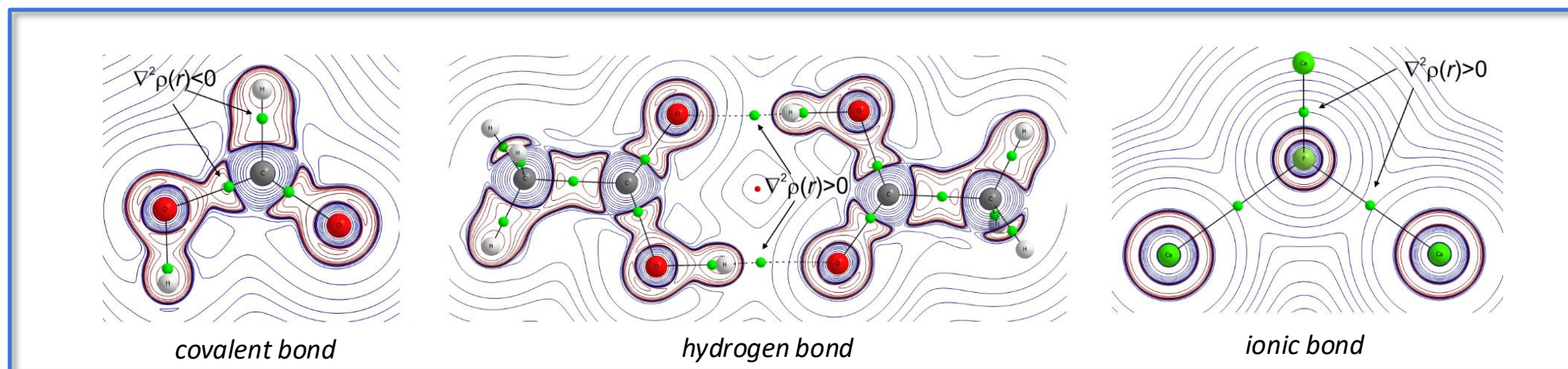
- Analysis of chemical bonding in a studied system – strenght and character
- Available from **both** experimental and theoretical data

electron density

Laplacian of electron density

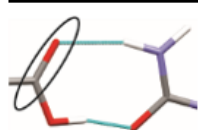
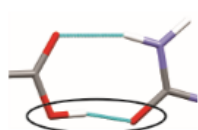
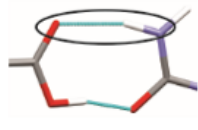
electron localization function (ELF)

potential and kinetic energy densities



TOPOND23 – Fully Integrated with CRYSTAL24 Code

Topological analysis of electron density

Examples	$\rho(r_{BCP})$	$\nabla^2\rho(r_{BCP})$	$V(r_{BCP})$	$G(r_{BCP})$	$H(r_{BCP})$	$ V(r_{BCP}) /G(r_{BCP})$	Type of interaction
	0.41	-0.67	--1.31	0.57	-0.74	2.3	Shared
	0.07	0.16	-0.07	0.06	-0.01	1.3	Intermediate
	0.02	0.07	-0.02	0.03	0.01	0.9	Closed-shell

$$\rho(r_{BCP}) - e/\text{Bohr}^3; \nabla^2\rho - e/\text{Bohr}^5; H(r_{BCP}), V(r_{BCP}), G(r_{BCP}) - \text{Hartree}$$

QTAIM – strategy

- ❑ Directly from wave function calculations – only atomic coordinates and space group needed
 - ✓ run single point calculations – electron density analysis does not require geometry optimization^(*)
 - ✓ Use either static or dynamic model
 - ⇒ for dynamic model provide ADPs (calculated within FREQCALC block)
 - ⇒ include DW part in *.d3 preparation
 - ✓ create *.d3 file – choose the right module
 - ⇒ TRHO module – topological analysis of electron density $\rho(\mathbf{r})$
 - ⇒ TLAP module – topological analysis of Laplacian of electron density $\nabla^2\rho(\mathbf{r})$
 - ⇒ ATBP module – atomic basin properties (best to be done once TRHO procedure is completed)

TOPO
TRHO
–2
0, 0, 0, 0, 20, 15, 10., 0.
1, 0., –0.5, –0.4047

TOPO
TLAP
0
0,0,1,0,20,7,5.
12,18
–

TOPO
ATBP
UNI
1,0.43
10,0,1,0

* You may consider refining hydrogen atoms positions (FRAGMENT keyword)

QTAIM – strategy

- ❑ Generation of theoretical structure factors
 - ✓ run single point calculations – you need a wave function
 - ✓ create *.d3 file using XFAC keyword
 - ⇒ use Miller indices hkl from X-ray diffraction experiment – ideal to compare experimental and theoretical data
 - ⇒ Generate Miller indices automatically – no dependence on Laue class
 - ✓ extract calculated structure factors
 - ✓ use external software to perform refinement – e.g. XD2016, MoPro etc.

```
RDFMWF
XFAC
9406 1
-13 0 1
-13 0 3
-13 0 5
-13 0 7
-13 0 9
-13 0 10
```

```
RDFMWF
XFAC
-15 1
END
```


QTAIM – strategy

❑ *.outp file – contains final data

TRHO

```

CP N.      132
*****

CP TYPE           : (3,-1)
COORD(AU) (X Y Z) : 9.1785E+00 3.7150E+00 4.7842E+00
COORD FRACT. CONV. CELL : -4.2790E-01 9.1285E-02 2.2863E-01
PROPERTIES (RHO,GRHO,LAP) : 3.7960E-01 7.2555E-15 -1.7097E+00
RHOA,SPIN DENSITY : 1.8991E-01 2.2301E-04
KINETIC ENERGY DENSITIES (G,K) : 8.5992E-02 5.1341E-01
VIRIAL DENSITY : -5.9940E-01
ELF(PAA) : 9.7774E-01
ELF(PBB) : 9.7797E-01

SPECTRAL DECOMP. OF THE HESSIAN OF RHO

EIGENVALUES (L1 L2 L3) : -1.1329E+00 -1.1309E+00 5.5408E-01
EIGENVECTORS : -9.9432E-01 1.0503E-01 1.7178E-02
               5.2997E-02 6.2864E-01 -7.7589E-01
               9.2288E-02 7.7057E-01 6.3064E-01

ELLIPTICITY : 1.8196E-03
    
```

ATBP

```

ATOMIC POPULATIONS

      N 1.70765043906408E+01
      Q -7.65043906407925E-02
      NA 8.53817733717140E+00
      NB 8.53832705744534E+00
      ESAP -1.49720273947551E-04

ATOMIC ENERGIES

      L -1.12594527768465E-03
      G 4.59050616516523E+02
      K 4.59049490288591E+02
      EG -4.76928715872815E+02
      EK -4.76927545783020E+02
      VNEO -1.09519908377694E+03
      VNEOC -1.11652579300050E+03

ATOMIC FORCES

      FAXA 1.22061209138622E-02
      FAYA 2.21797599823615E-01
      FAZA 3.31817091496728E-01
    
```

XFAC

```

*****
*                                     *
*               X-RAY STATIC STRUCTURE FACTORS               *
*                                     *
*****

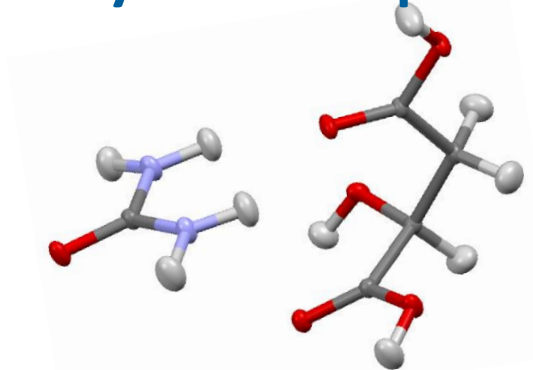
NUMBER OF FACTORS COMPUTED :19182

      ALPHA+BETA ELECTRONS
      H K L PX PY PZ REAL PART IMAG. PART Fhkl

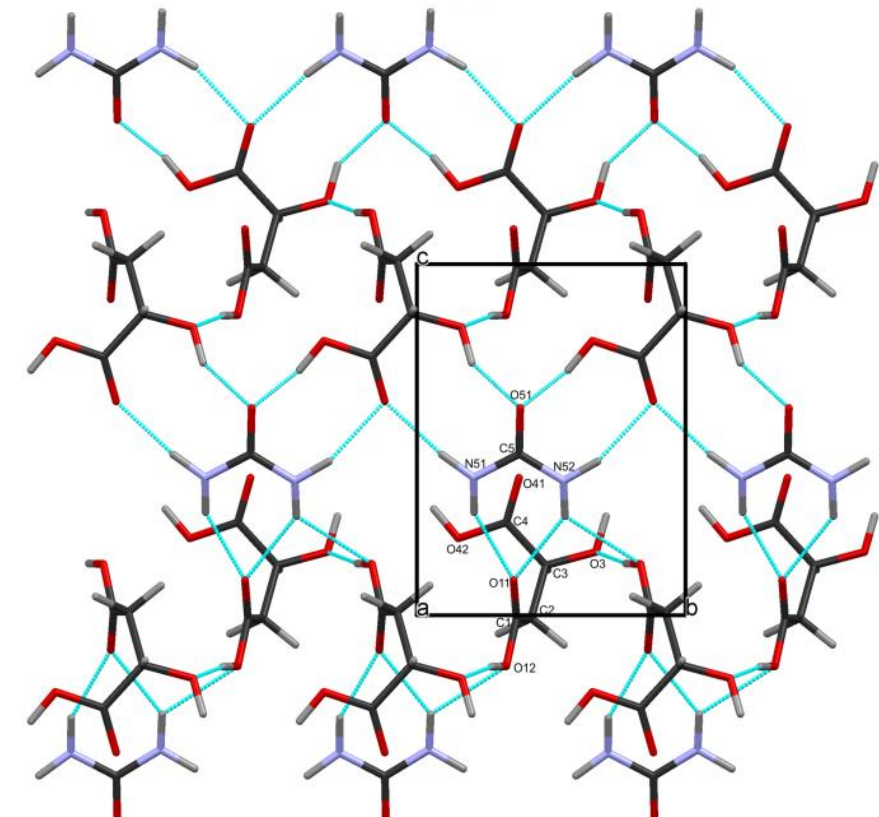
1 0 0 0.237 0.000 0.118 0.1145E+02 0.5329E-14 0.1145E+02
2 0 0 0.475 0.000 0.235 0.7188E+01 0.3553E-14 0.7188E+01
3 0 0 0.712 0.000 0.353 0.4635E+02 0.1421E-13 0.4635E+02
4 0 0 0.949 0.000 0.470 0.9764E+02 -0.2043E-13 0.9764E+02
5 0 0 1.186 0.000 0.588 0.1031E+03 -0.2787E-13 0.1031E+03
6 0 0 1.424 0.000 0.705 -0.7927E+02 0.6289E-13 0.7927E+02
7 0 0 1.661 0.000 0.823 0.8343E+02 -0.5729E-13 0.8343E+02
8 0 0 1.898 0.000 0.941 0.4714E+02 0.4891E-13 0.4714E+02
9 0 0 2.135 0.000 1.058 0.7093E+02 -0.9193E-13 0.7093E+02
10 0 0 2.373 0.000 1.176 -0.1085E+03 0.5751E-13 0.1085E+03
11 0 0 2.610 0.000 1.293 0.3223E+02 0.6856E-14 0.3223E+02
12 0 0 2.847 0.000 1.411 -0.4741E+02 -0.3153E-13 0.4741E+02
13 0 0 3.085 0.000 1.529 -0.2785E+02 0.4396E-13 0.2785E+02
14 0 0 3.322 0.000 1.646 -0.4175E+02 -0.7616E-13 0.4175E+02
15 0 0 3.559 0.000 1.764 -0.9241E+02 0.1162E-12 0.9241E+02
16 0 0 3.796 0.000 1.881 0.5607E+02 -0.9104E-13 0.5607E+02
17 0 0 4.034 0.000 1.999 -0.8643E+02 0.6833E-13 0.8643E+02
    
```

Electron density – theory vs. experiment

- ❑ Co-crystal of L-malic acid and urea
- ❑ Efficient NLO material (ULMA)
- ❑ Computational details
 - ✓ no geometry optimization
 - ✓ theoretical structure factors generated for experimental Miller indices – resolution 1.13 Å⁻¹

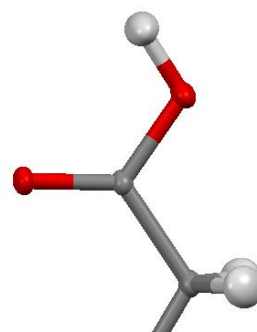
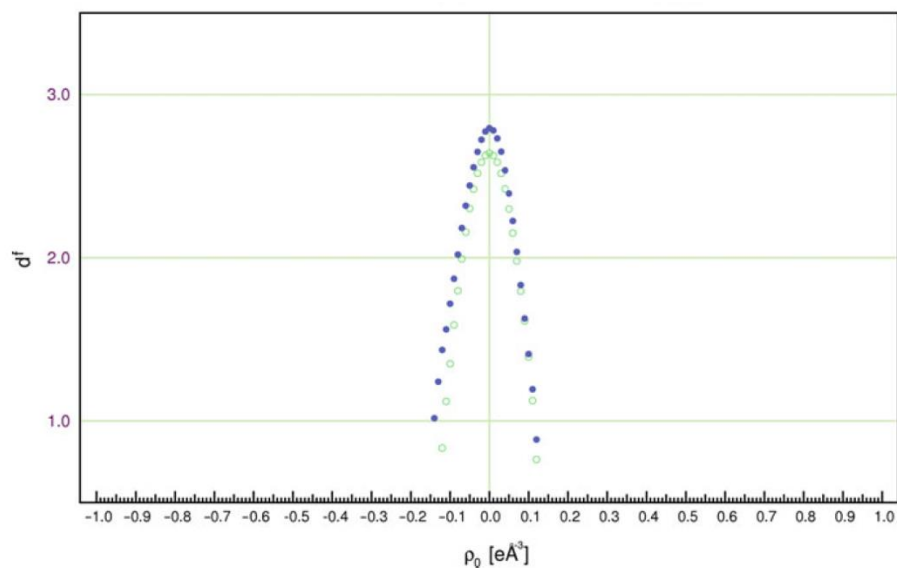


a, b, c / Å	6.6411(2) 6.9162(2) 9.0495(2)
β/°	94.697(1)
Unit cell volume/Å ³	414.26(2)
Temperature/K	80(2)
Space group	P2 ₁
Z	2



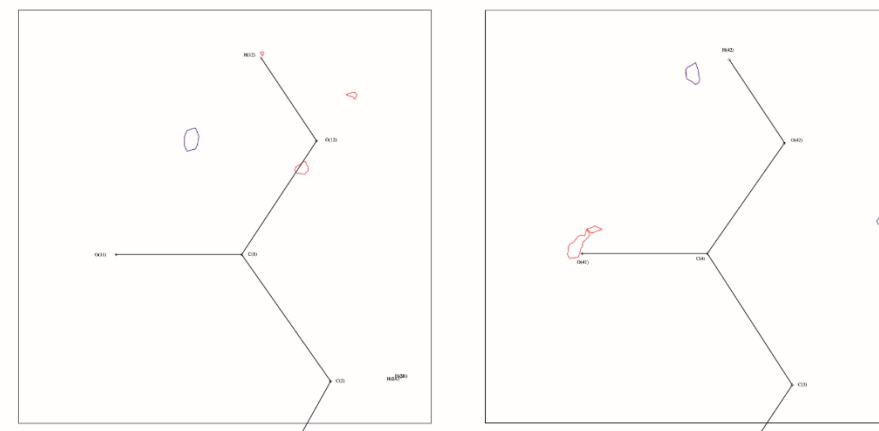
Electron density – theory vs. experiment

Fractal dimensions vs electron density

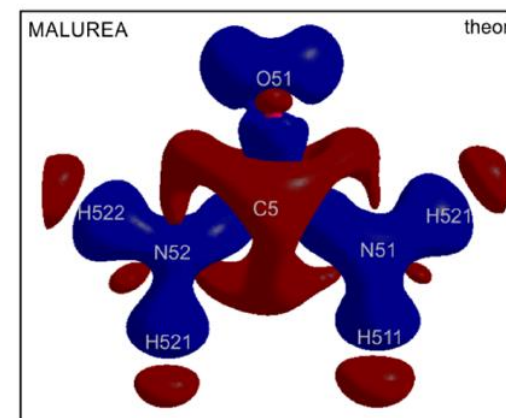
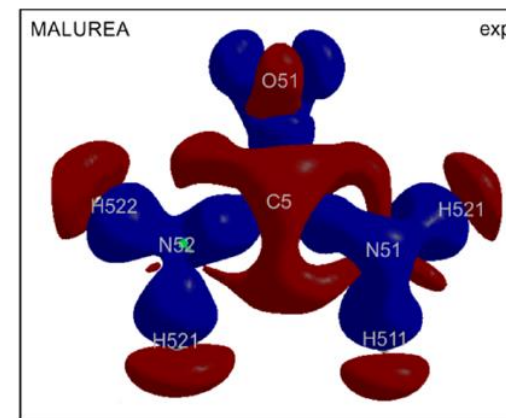
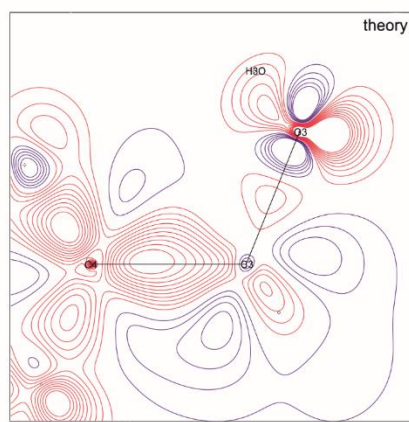
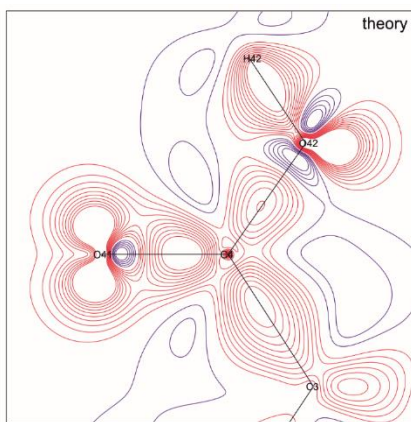
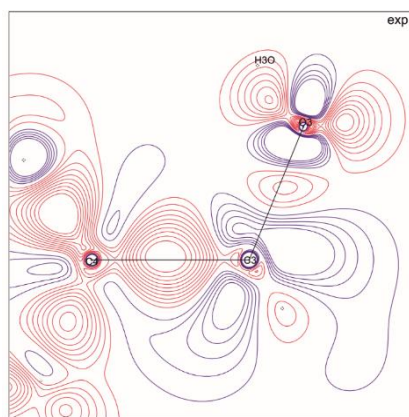
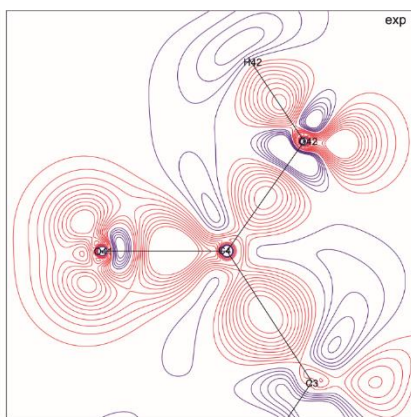
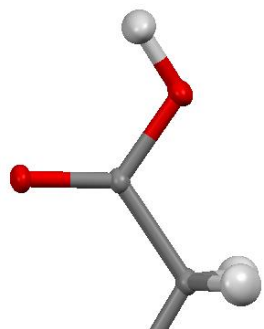


model	d^f	e_{gross} e	ρ_{min} $\text{e}\text{\AA}^{-3}$	ρ_{max} $\text{e}\text{\AA}^{-3}$	$\Delta\rho$ $\text{e}\text{\AA}^{-3}$
Exp	2.83	4.86	-0.164	0.156	0.034
Theory	2.76	4.63	-0.153	0.180	0.027

Residual density maps, contour spacing is $0.1\text{\AA}^{-1}$



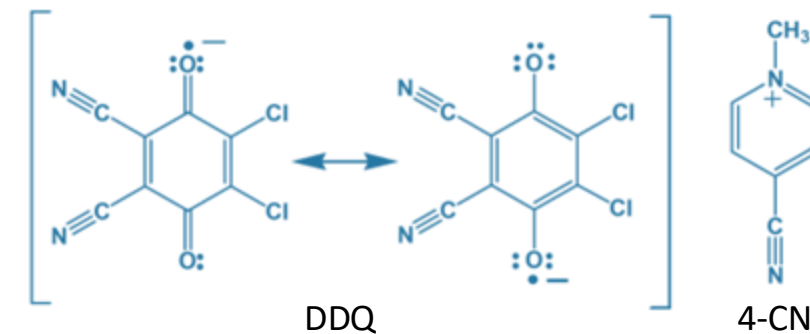
Electron density – theory vs. experiment



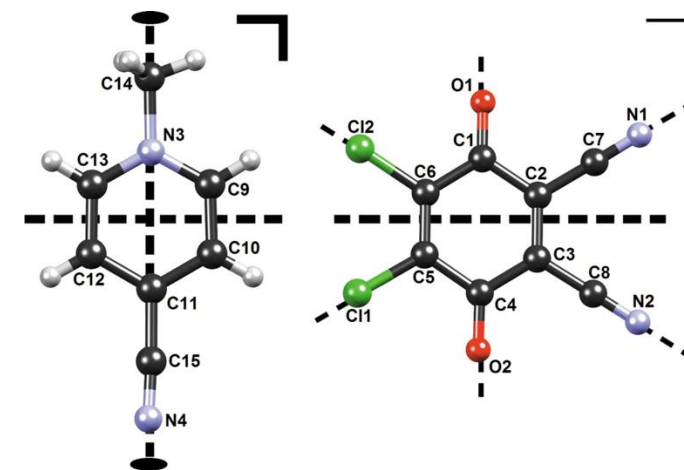
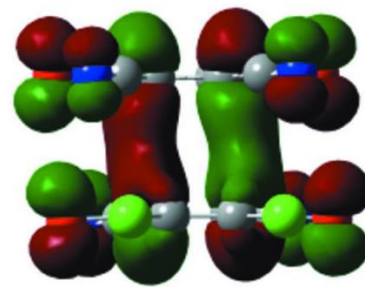
2D and 3D deformation electron density

Chemical bonding vs phase transition

What happens with intermolecular interactions between radical anions upon phase transition?

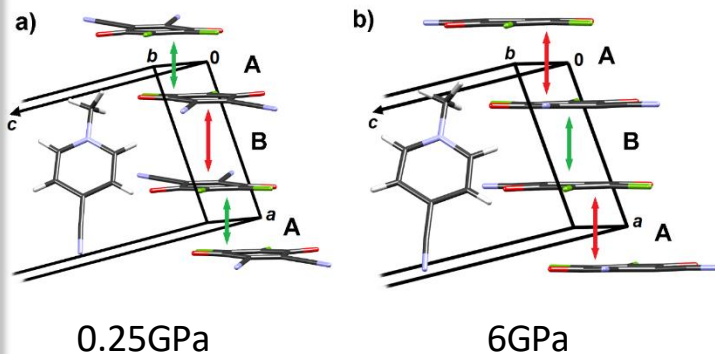


- ❑ Salt of 5,6-dichloro-2,3-dicyanosemiquinone radical anion (DDQ) with 4-cyano-N-methylpyridinium cation (4-CN)
 - ✓ presence of pancake bonding
 - ✓ space group $P2_1/n$
 - ✓ studies under high pressure: 0.25 GPa – 6 GPa
 - ✓ no high-resolution data available for high pressure structures
 - ✓ need for periodic calculations to assess electron density distribution

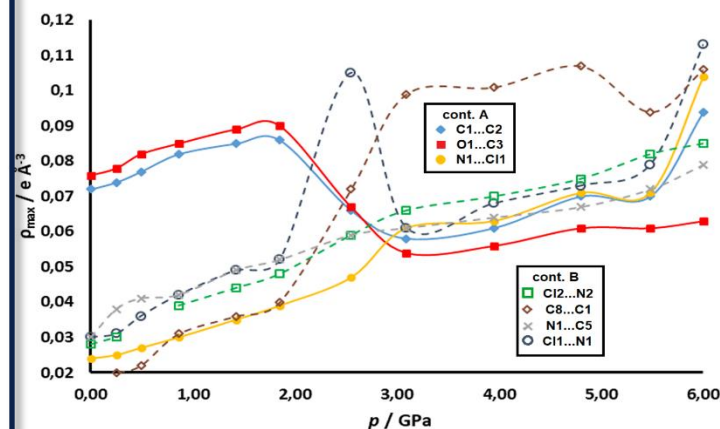


Chemical bonding vs. phase transition

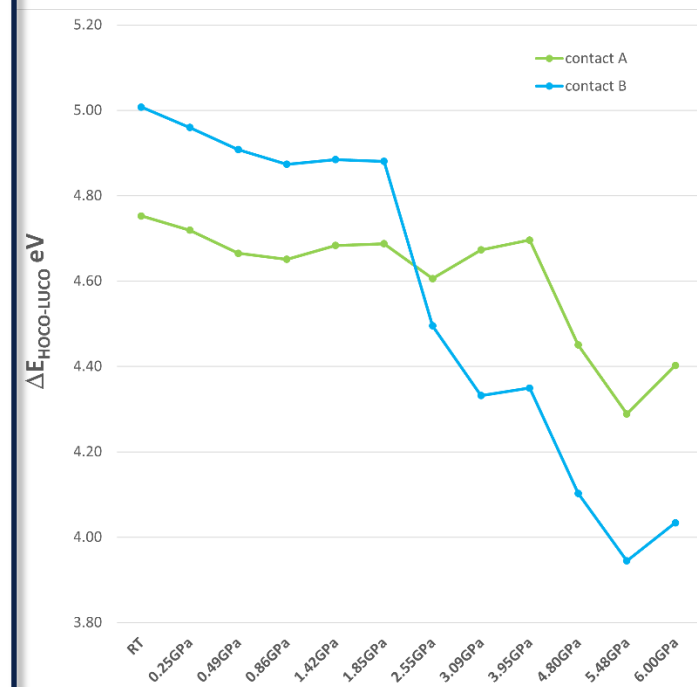
Crystal packing



Electron density at the BCP



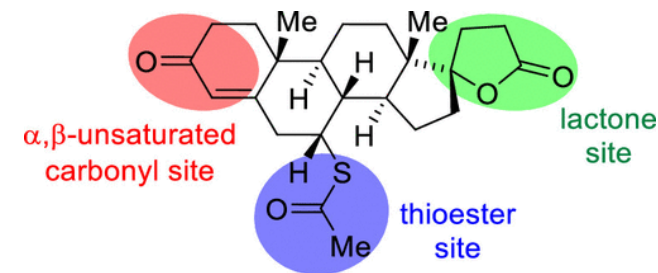
HOCO-LUCO energy



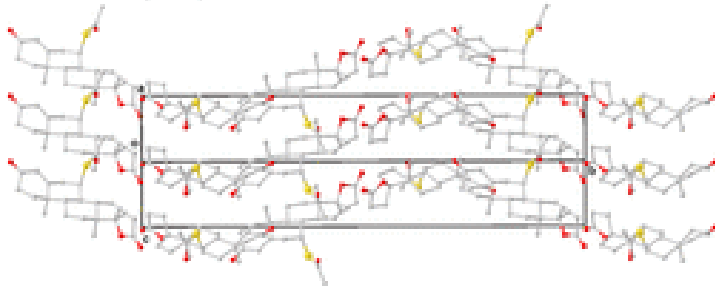
Studies of polymorphism

❑ Spironolactone molecule

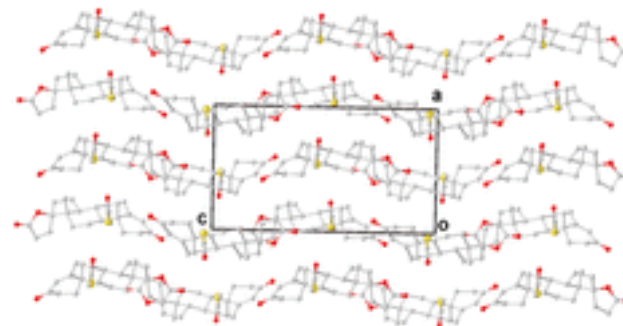
- ✓ active pharmaceutical ingredient API – steroidal drug
- ✓ form II used in commercially available pills
- ✓ crystallizes in three polymorphic forms
- ✓ no access to high-resolution data - poor quality crystals
- ✓ stability of the polymorphs studied



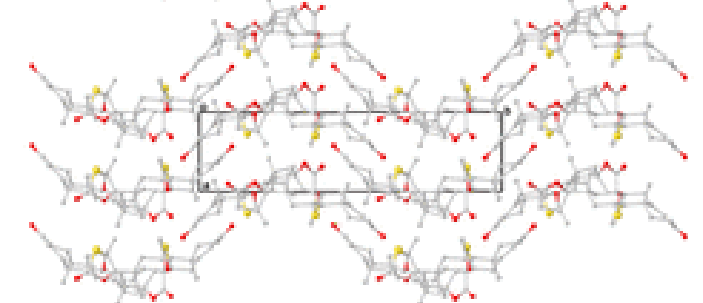
I, 295 K, (101)



II, 298 K, (010)



III, 100 K, (001)



Studies of polymorphism

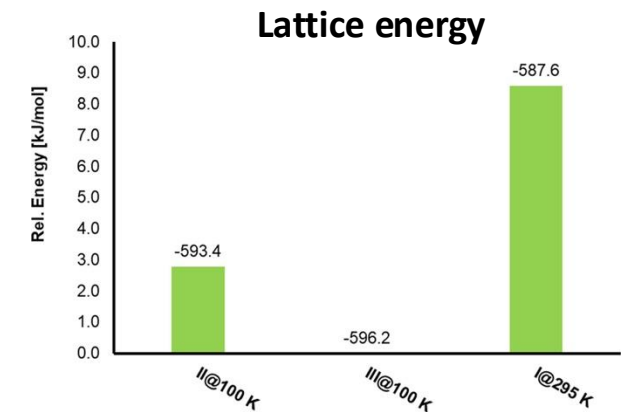
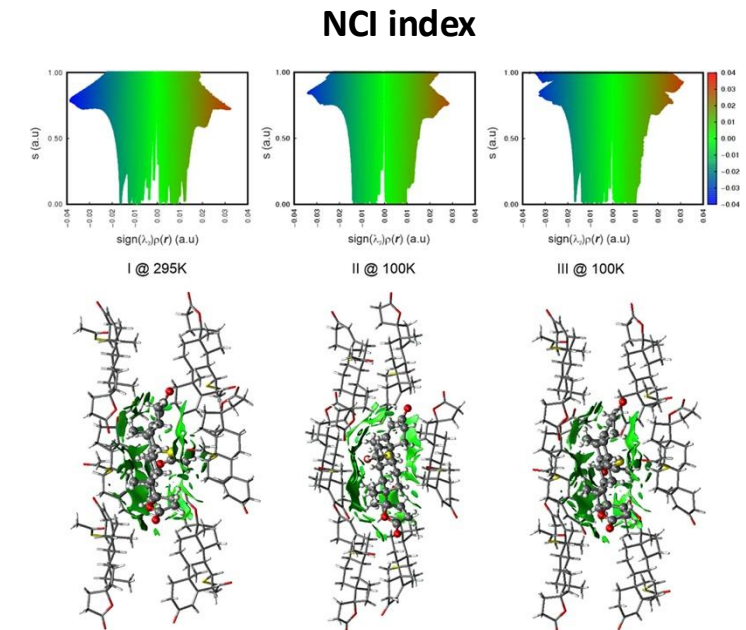
□ Lattice energies calculations

$$\Delta E = \left(\frac{E_{cryst}}{Z} \right) - \left(\sum_i^N E_{i mol} \right) + BSSE$$

- ⇒ total energy E_{cryst} of the crystal including dispersion correction
- ⇒ energy of a single molecule $E_{i mol}$ – extract molecule from molecular crystal
- ⇒ BSSE by using Boys-Bernardi counterpoise method

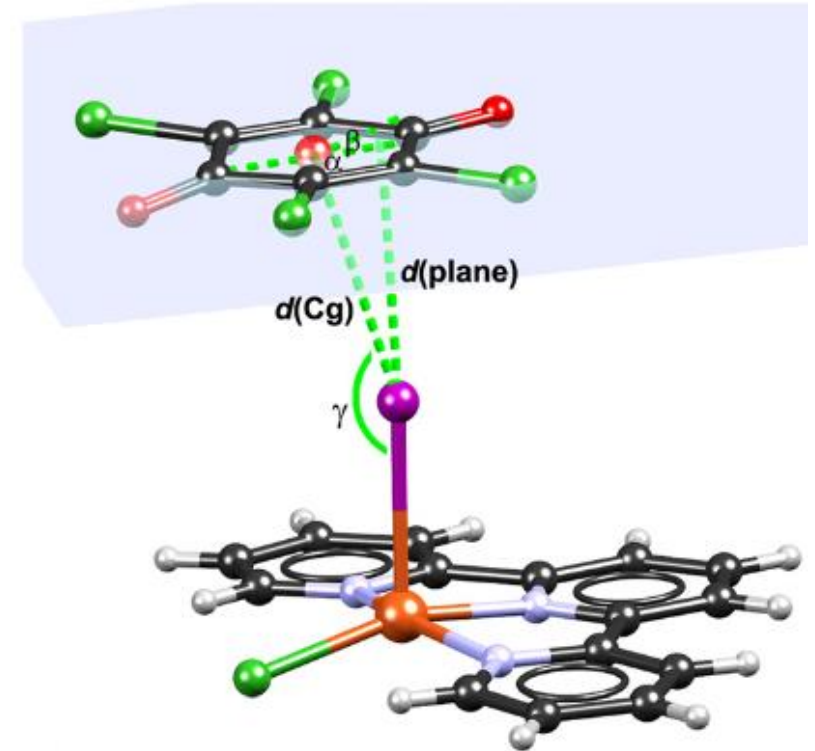
□ Form III found to be most stable – not expected

- ✓ disorder present in experimental data not included in calculations
- ✓ strength of interactions indicates form II to be most stable with the strongest intermolecular interactions



Structure-property correlation

- ❑ Metal-bound halide and a quinoid ring cocrystal: $[\text{Cu}(\text{terpy})\text{ClI}] \cdot \text{X}'_4\text{Q}$ (terpy = 2,2':6',2''-terpyridine; Q = quinone)
 - ✓ is the presence of π -hole interactions responsible for bulk conductivity?
 - ✓ is the charge transfer only between iodine and quinone ring?
- ❑ Periodic calculations engaged:
 - ✓ crystal orbital insight
 - ✓ band structure and density of state (DOS) assessment



Band gap and DOS – strategy

- ❑ Optimize geometry
- ❑ Run single point calculations – you need a wave function
- ❑ Prepare *.d3 file
 - ✓ BWIDTH – band range and band gap value
 - ✓ BAND – k-path definition and band structure calculations
 - ✓ DOS (density of state):
 - ⇒ summary of electronic structure of the material
 - ⇒ information on the number and character of one-electron levels (molecular orbitals) as a function of the (orbital) energy

```
NEWK
0 5
5 5 5
1 0
BWIDTH
0 0
END
```

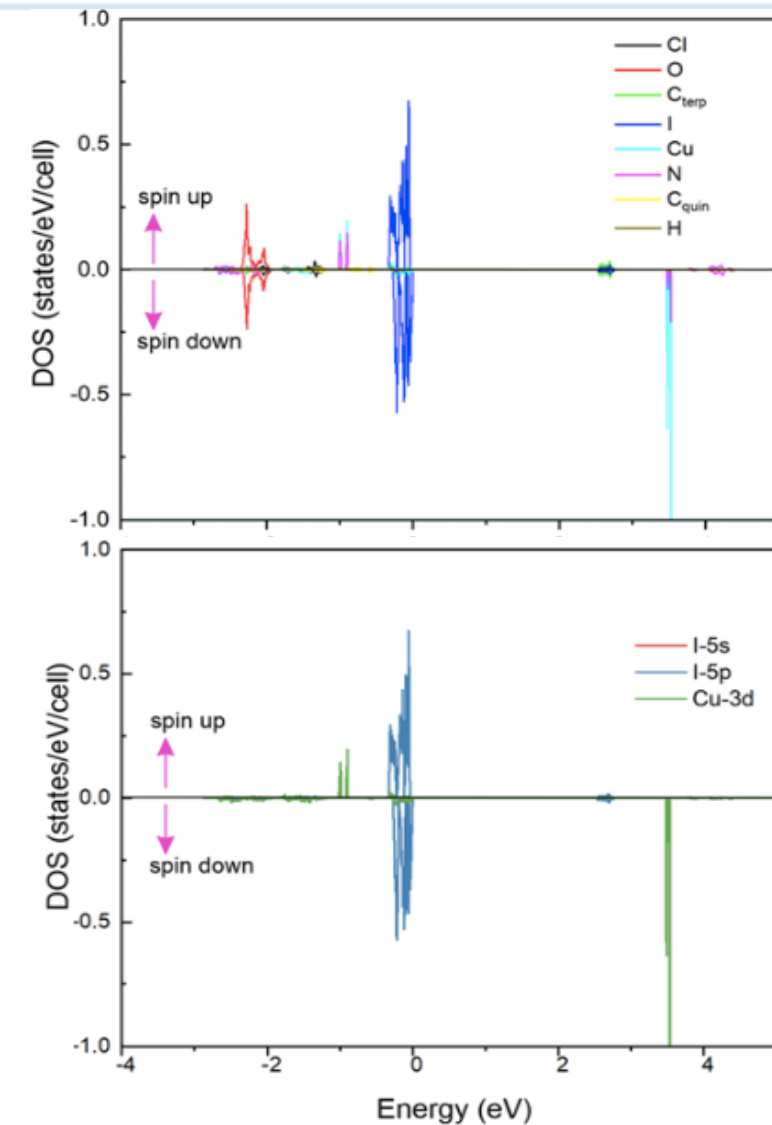
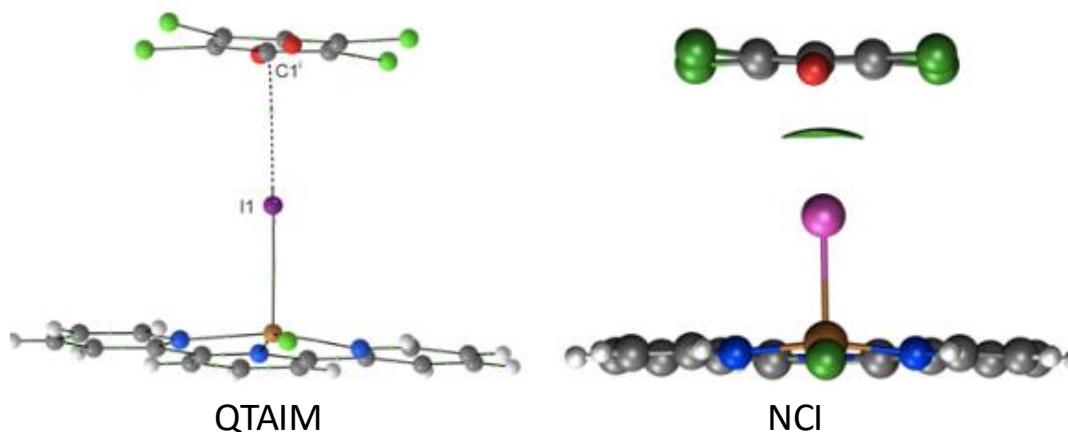
```
BAND
form I P21/c path:
10 4 100 500 600 1 0
0 0 0 0 2 0
0 2 0 0 2 2
0 2 2 0 0 2
0 0 2 0 0 0
0 0 0 -2 0 2
-2 0 2 -2 2 2
-2 2 2 0 2 0
0 2 0 -2 2 0
-2 2 0 -2 0 0
-2 0 0 0 0 0
END
```

```
NEWK
5 10
1 0
DOSS
3 6002 570 650 1 12 0
3 737 738 739 All s-orbitals of I no 493
9 740 741 742 743 744 745 746 747 748
10 859 860 861 862 863 864 865 866 867 868
END
```

Structure-property correlation

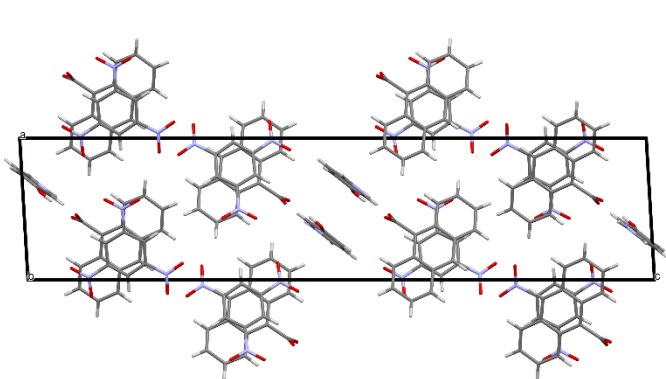
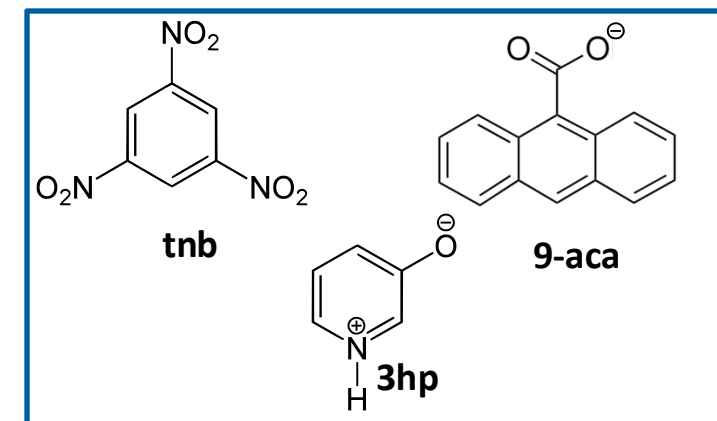
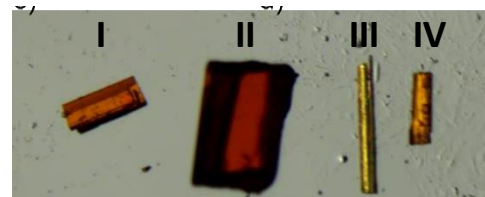
□ QTAIM and DOS analysis

- ✓ directional attractive character of I... π interaction
- ✓ dominant contribution to the top of the valence band is attributed to the iodine atom – p-orbitals
- ✓ minor contribution from d-d transitions of the copper ion

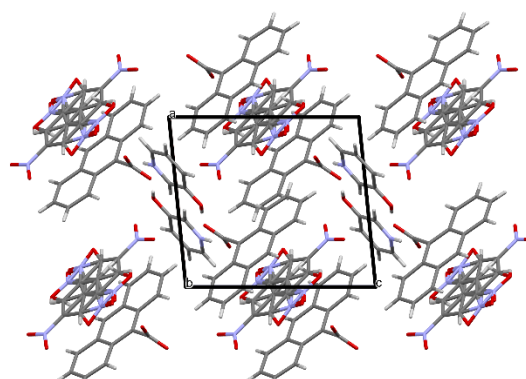


Structure-property correlation

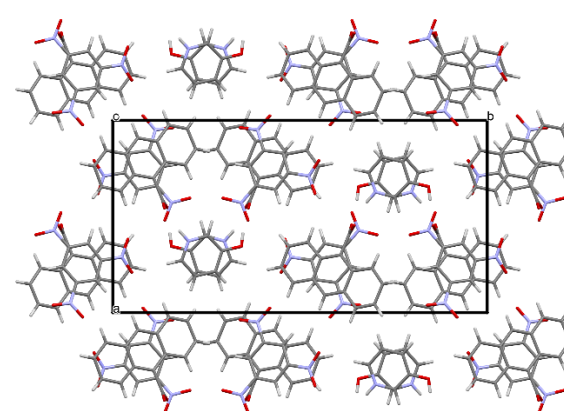
- ❑ Polymorphs of multi-component molecular adducts
 - ✓ 4 distinctive polymorphic forms
 - ✓ same types of intermolecular interactions present:
 - $\text{O}-\text{H}\cdots\text{O}^{(-)}$ and $\text{N}^{(+)}-\text{H}\cdots\text{O}^{(-)}$
 - ✓ designed via crystal engineering tools



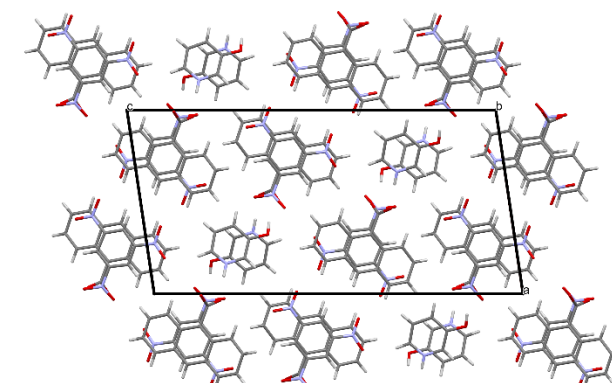
Form I
 $P2_1/c$



Form II
 $P\bar{1}$



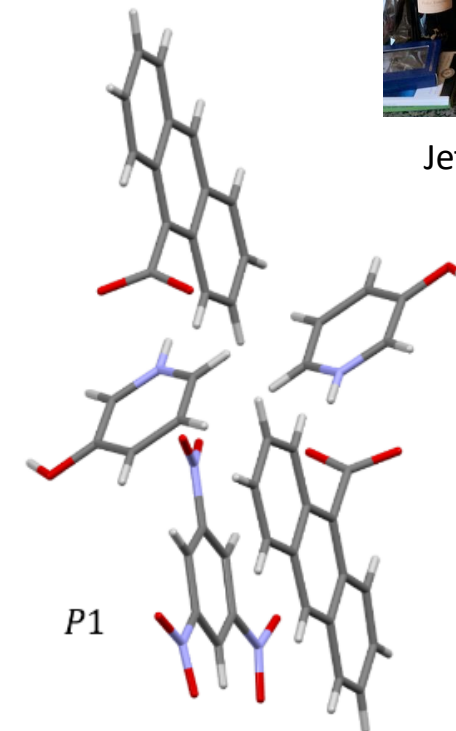
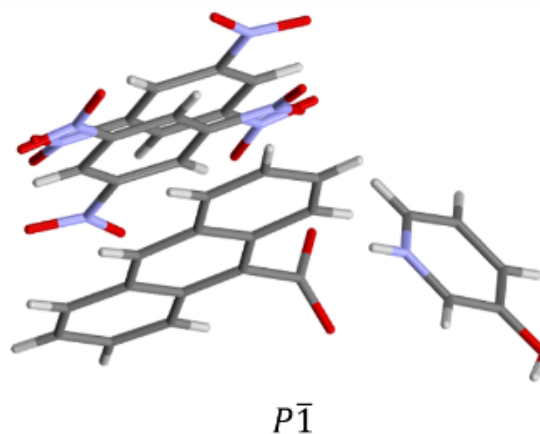
Form III
 $P2_1/c$



Form IV
 $P2_1/n$

Structure-property correlation

- ❑ Form II shows disorder
 - ✓ trinitrobenzene (**tnb**) lies on a center of inversion and occupying two different orientations – occupation 0.5
 - ✓ necessity to “modify” the crystal structure
 - ✓ two separate files with two different orientations of **tnb** – averaged values taken as a final result

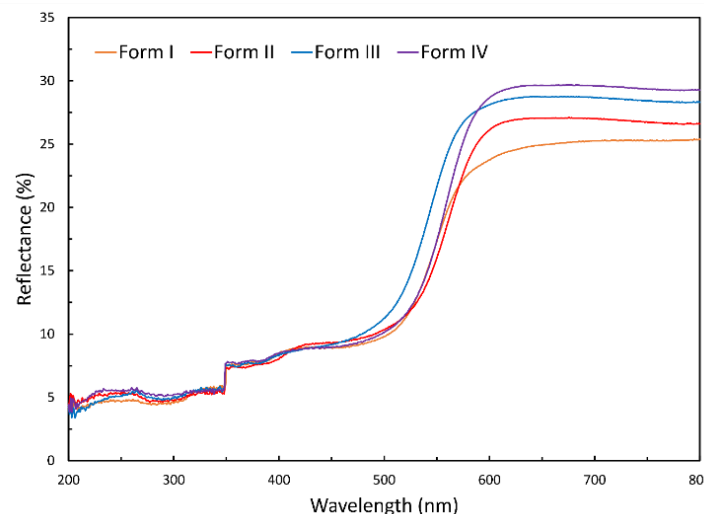


Jefferson de Andrade
EDITOR-IN-CHF

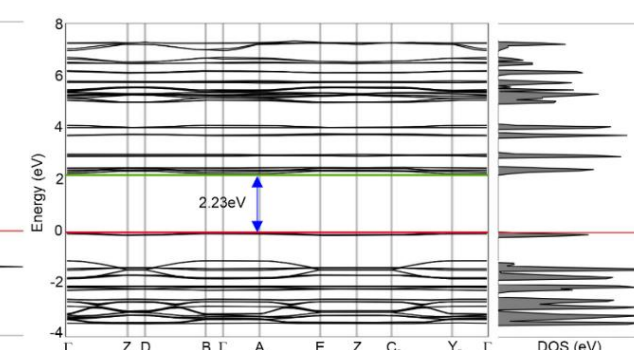
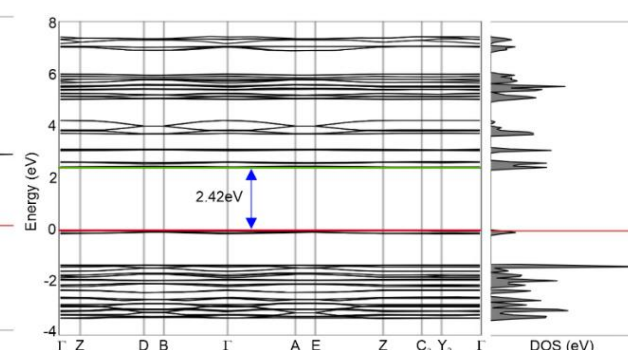
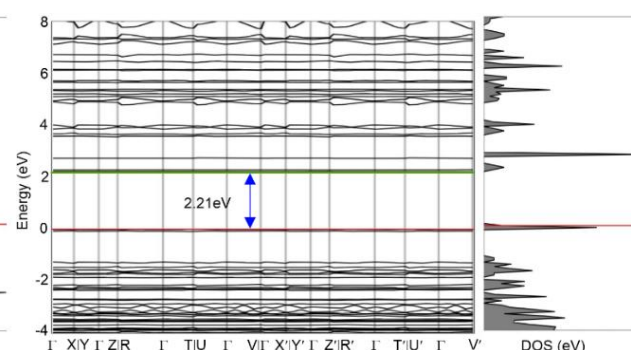
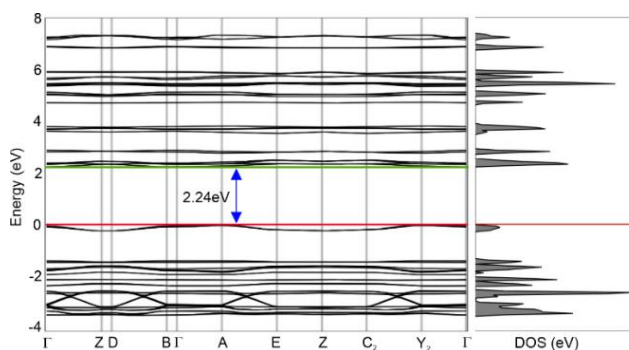
Structure-property correlation

Comparison between polymorphs:

- ✓ UV-VIS via diffuse reflectance spectroscopy – experimental optical band gap estimation
- ✓ Band structure calculations



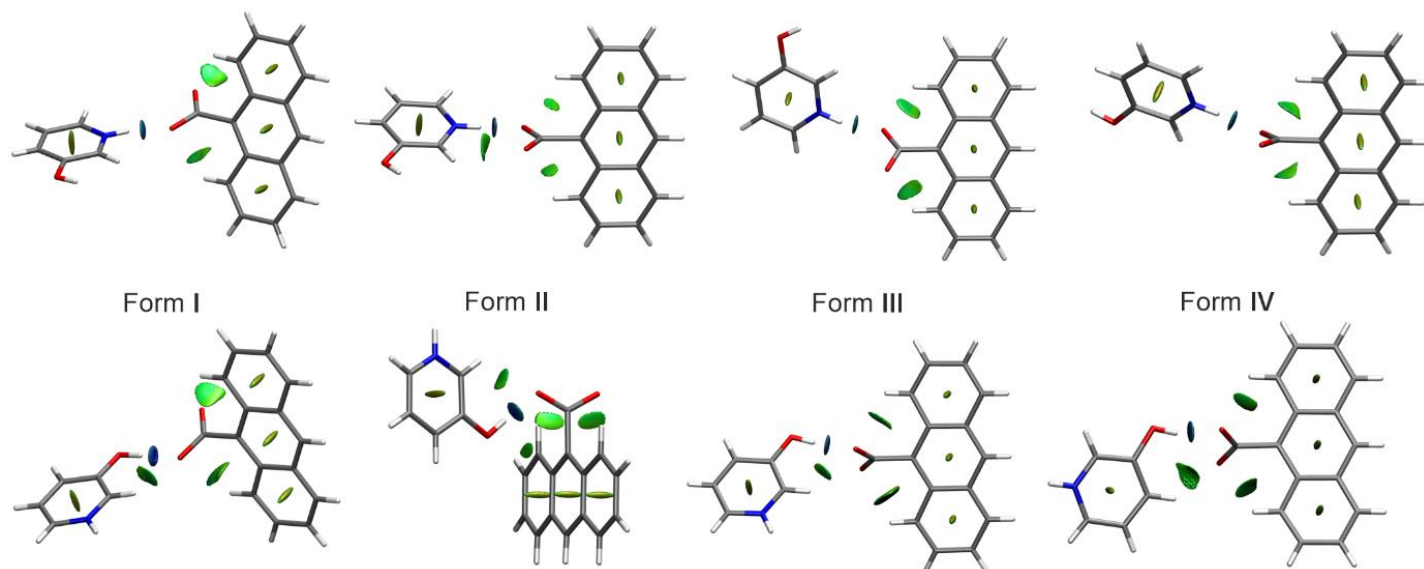
	Experimental / eV	Periodic DFT / eV
Form I	1.900	2.24
Form II	1.876	2.21
Form III	1.957	2.42
Form IV	1.912	2.23



Structure-property correlation

Comparison between polymorphs:

✓ intermolecular interactions with QTAIM



	$\rho(\mathbf{r})$ /eÅ ³	$\nabla^2\rho(\mathbf{r})$ /eÅ ⁵	$G(\mathbf{r})$	$V(\mathbf{r})$	$E(\mathbf{r})$	$ V(\mathbf{r}) $ / $G(\mathbf{r})$
O9–H9...O7						
I	0.533	4.58	0.42	-0.52	-0.10	1.23
II	0.466	4.58	0.38	-0.44	-0.06	1.17
III	0.421	4.10	0.34	-0.39	-0.05	1.15
IV	0.435	4.34	0.36	-0.41	-0.05	1.14
N4–H4A...O8						
I	0.362	4.10	0.31	-0.34	-0.02	1.08
II	0.464	4.34	0.37	-0.43	-0.06	1.17
III	0.400	4.34	0.34	-0.37	-0.03	1.10
IV	0.383	4.10	0.32	-0.35	-0.03	1.11

Physical properties prediction – linear optical properties



POLARIZABILITY

SUSCEPTIBILITY

DIELECTRIC
CONSTANT

REFRACTIVE
INDEX

$$\begin{pmatrix} \alpha_{11} & \alpha_{12} & \alpha_{13} \\ \alpha_{12} & \alpha_{22} & \alpha_{23} \\ \alpha_{13} & \alpha_{23} & \alpha_{33} \end{pmatrix}$$

$$\begin{pmatrix} \chi_{11} & \chi_{12} & \chi_{13} \\ \chi_{12} & \chi_{22} & \chi_{23} \\ \chi_{13} & \chi_{23} & \chi_{33} \end{pmatrix}$$

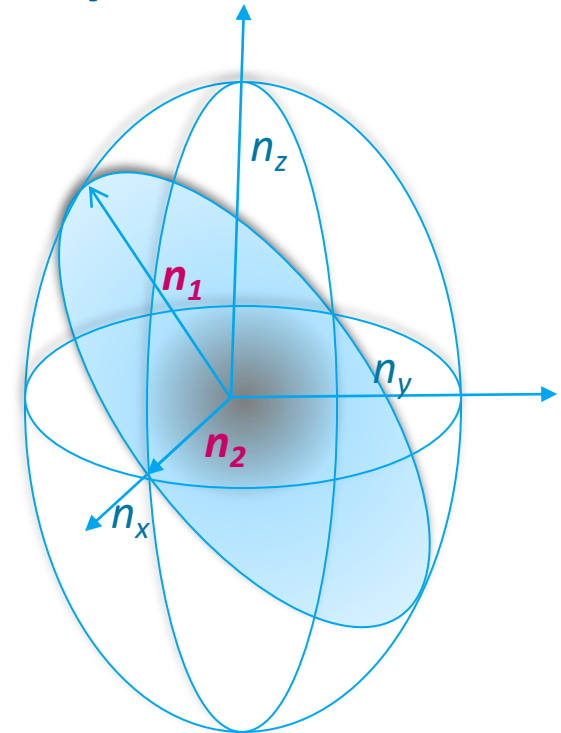
$$\begin{pmatrix} \epsilon_{11} & \epsilon_{12} & \epsilon_{13} \\ \epsilon_{12} & \epsilon_{22} & \epsilon_{23} \\ \epsilon_{13} & \epsilon_{23} & \epsilon_{33} \end{pmatrix}$$

$$\begin{pmatrix} \sqrt{\epsilon_{11}} & 0 & 0 \\ 0 & \sqrt{\epsilon_{22}} & 0 \\ 0 & 0 & \sqrt{\epsilon_{33}} \end{pmatrix}$$

$$\chi_{ij}^{(1)} = \frac{\alpha_{ij}}{V}$$

$$\epsilon_{ij} = \chi_{ij}^{(1)} + \delta_{ij}$$

$$n_{ii} = \sqrt{\epsilon_{ii}}$$

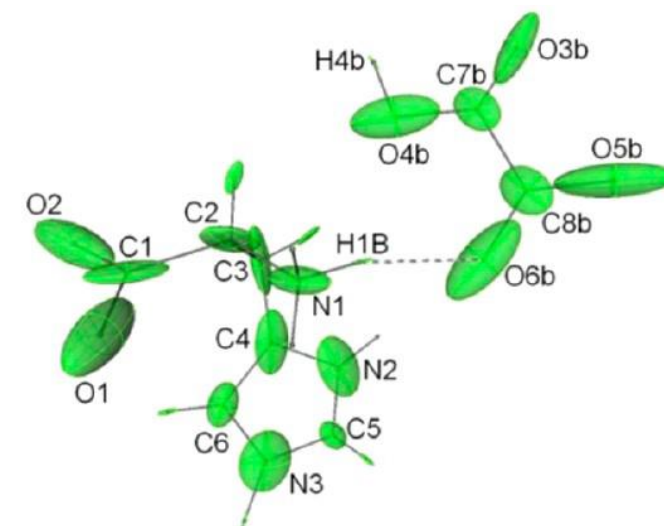
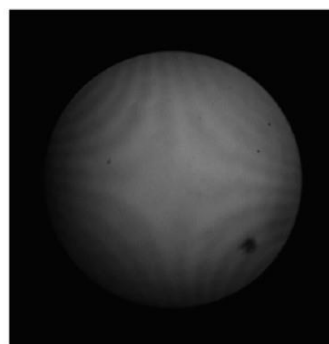
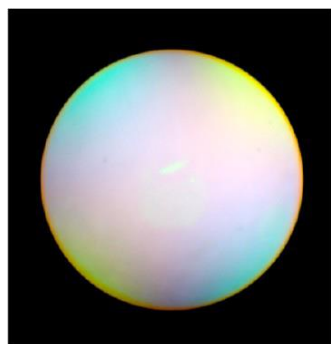
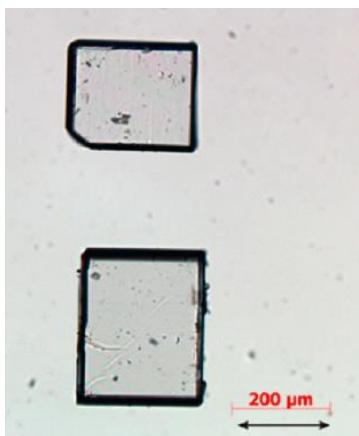
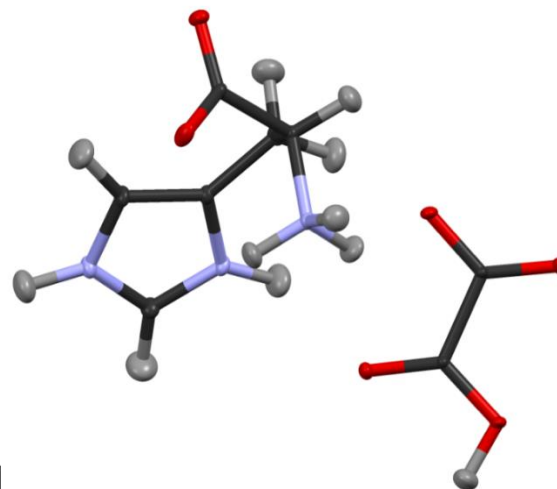


$$n_i = \sqrt{\epsilon_{ii}}$$

Structure-refractive index correlation

□ L-histidinium hydrogen oxalate

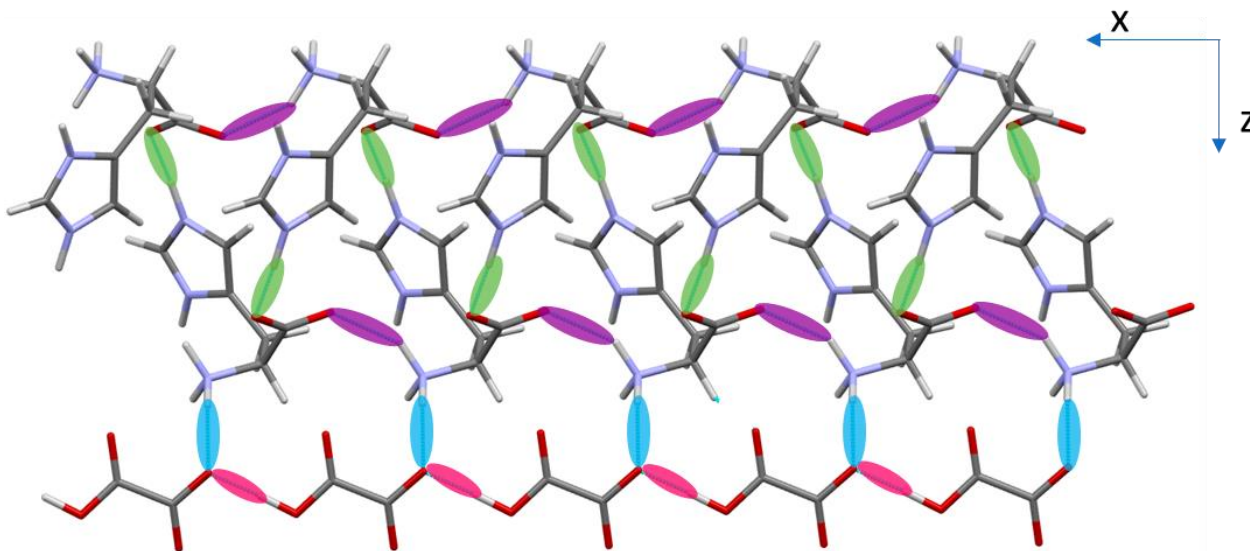
- ✓ optically biaxial negative
- ✓ space group $P2_12_12_1$
- ✓ potential organic NLO material



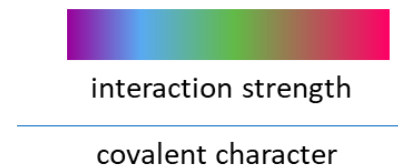
Structure-refractive index correlation

□ QTAIM vs. refractive index calculations:

- ✓ Strongest interactions along X determine the highest refractive index
- ✓ No strong interactions along Y axis – atomic/molecular polarizability tensors have their smallest components, thus smallest refractive index

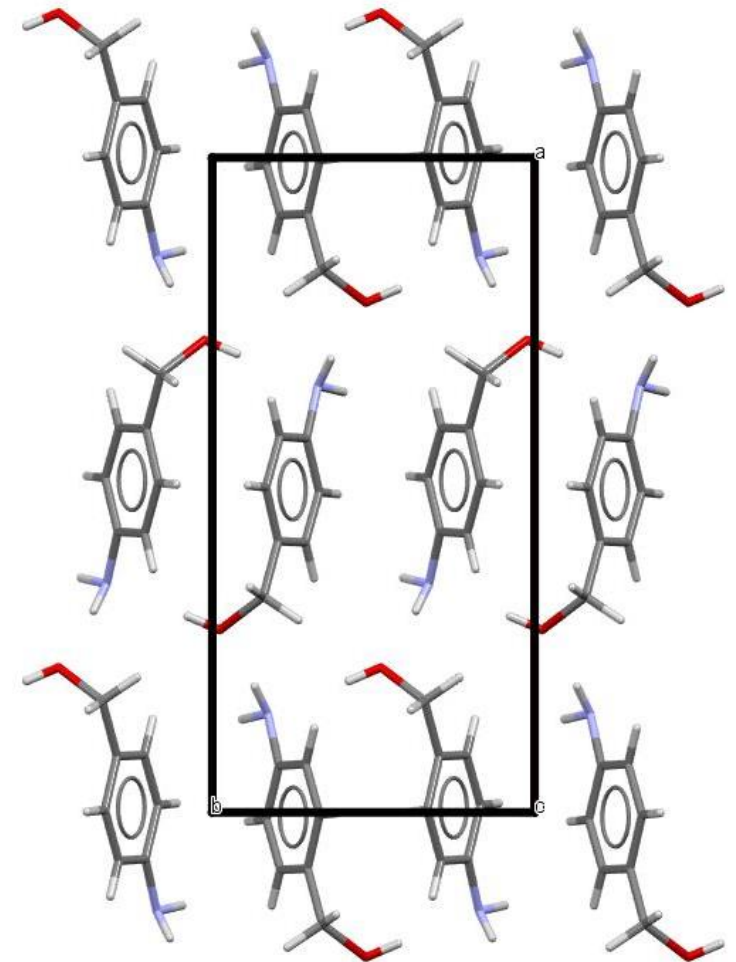
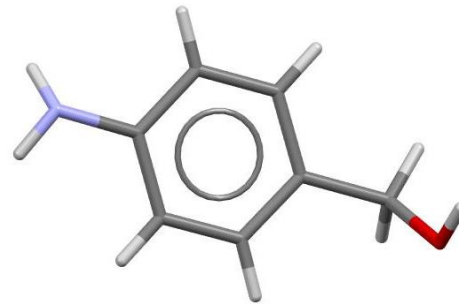


	n_1	n_2	n_3
Experiment	1.63	1.50	1.60
PDFT (B3LYP)	1.58	1.45	1.63



Predicting linear optical properties

- 4-aminophenyl methanol
 - ✓ push-pull chromophore
 - ✓ polar space group



a, b, c/Å	15.0879(7), 7.4306(3), 5.7751(1)
Z	4
Space group	<i>Pna21</i>

Predicting linear optical properties

```
*****
POLARIZABILITY (ALPHA), DIELECTRIC (EPSILON) AND FIRST-ORDER ELECTRIC
SUSCEPTIBILITY (CHI(1)) TENSORS (INDEPENDENT COMPONENTS)

FR. (eV) COMP.   ALPHA(Re,Im)      EPSILON(Re,Im)      CHI(1)
0.000   XX      587.0707      0.0000      2.7688      0.0000      1.7688
0.000   YY      407.3938      0.0000      2.2275      0.0000      1.2275
0.000   ZZ      521.7227      0.0000      2.5719      0.0000      1.5719

DIELECTRIC TENSOR

TENSOR IN PRINCIPAL AXES SYSTEM
AA  2.768821E+00 BB  2.227462E+00 CC  2.571930E+00

REFRACTIVE INDICES
AA  1.492468E+00 BB  1.603724E+00 CC  1.663978E+00

OPTICALLY ANISOTROPIC - BIAXIAL CRYSTAL

ACUTE ANGLE BETWEEN OPTIC AXES (DEGREES)
2V= 68.28266
BIAXIAL NEGATIVE: 2V ANGLE BISECTED BY THE SMALLEST REFRACTIVE INDEX (AA)

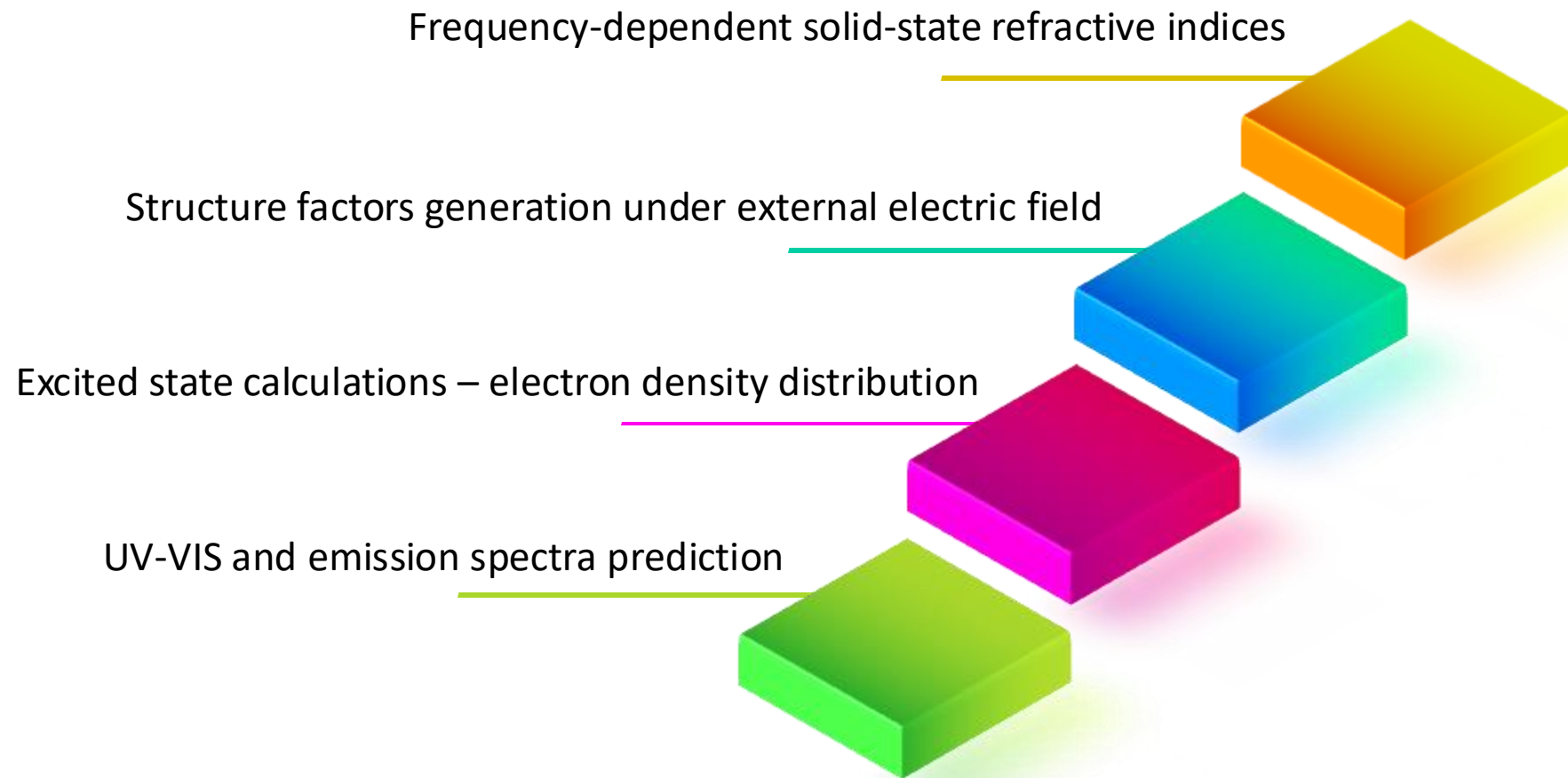
ANGLES BETWEEN OPTICAL AND BASIS LATTICE VECTORS (DEGREES)
      A1      A2      A3
O.A. 1  124.1413  34.1413  90.0000
O.A. 2  124.1413  145.8587  90.0000

EPSILON=N**2=1+CHI(1)=1+ALPHA*4*PI/V,  4*PI/V= 0.00301 BOHR**(-3)
ALPHA IN BOHR**3; EPSILON AND CHI(1) ARE DIMENSIONLESS.
*****
```

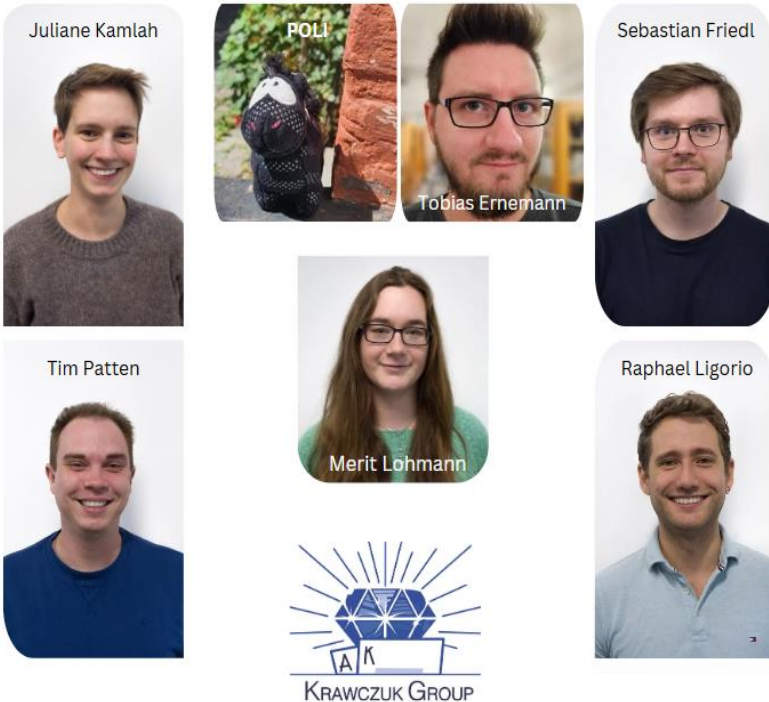
$$\overline{n_{exp}} = 1.538$$

$$\overline{n_{theo}} = 1.587$$

Perspective – what else?



Acknowledgement



- ❖ Nico Graw, University of Göttingen
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Thank you for your attention

